

## QAANAAQ AP, Greenland

WMO: 042050

Lat: 77.486N

Lon: 69.376W

Elev: 16

StdP: 101.13

Time Zone: -4.00 (NAA)

Period: 96-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -31.7      | -30.2 | -38.0                         | 0.1 | -30.5 | -36.3 | 0.1 | -29.1 | 22.4                  | -18.0 | 12.9 | -9.6 | 1.0                   | 160  | 0.977 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                     | (p)  |
| 7             | 5.1                       | 12.8            | 7.5  | 11.2 | 6.9  | 10.1 | 6.5  | 8.3                 | 11.1 | 7.6 | 10.3 | 6.9 | 9.4  | 4.9                     | 80   |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 6.5                             | 6.0 | 8.7  | 5.7 | 5.7 | 8.3  | 4.8 | 5.4 | 7.8  | 25.4          | 11.3 | 24.0 | 10.4 | 22.5 | 9.6  | 12.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| (4) 12.8          | 10.1 | 8.2 | -33.1                      | 15.6 | 2.6                | 2.0 | -35.0                                              | 17.0 | -36.5      | 18.2 | -37.9      | 19.3 | -39.8      | 20.8 |  |  |
| (5)               |      |     | -33.6                      | 9.0  | 2.5                | 1.2 | -35.4                                              | 9.8  | -36.9      | 10.5 | -38.3      | 11.2 | -40.1      | 12.1 |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct  | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)  | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | -8.6   | -21.2 | -22.5 | -21.8 | -14.1 | -3.9  | 3.2   | 6.7   | 5.6   | 0.0   | -6.7 | -12.3 | -17.2 |
|                                                                                  | DBStd       | 11.32  | 5.29  | 5.47  | 5.29  | 5.00  | 4.08  | 2.67  | 2.68  | 2.70  | 2.91  | 4.01 | 4.37  | 5.47  |
|                                                                                  | HDD10.0     | 6796   | 968   | 909   | 984   | 722   | 429   | 206   | 107   | 136   | 302   | 518  | 671   | 843   |
|                                                                                  | HDD18.3     | 9832   | 1227  | 1143  | 1243  | 972   | 688   | 455   | 361   | 394   | 552   | 777  | 921   | 1102  |
|                                                                                  | CDD10.0     | 6      | 0     | 0     | 0     | 0     | 0     | 1     | 4     | 1     | 0     | 0    | 0     | 0     |
|                                                                                  | CDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0     | 0     |
|                                                                                  | CDH23.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0     | 0     |
|                                                                                  | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0     | 0     |
| Wind                                                                             | WSAvg       | 2.3    | 1.6   | 1.7   | 1.5   | 1.6   | 2.0   | 2.4   | 2.7   | 2.8   | 3.1   | 3.3  | 3.0   | 2.0   |
| Precipitation                                                                    | PrecAvg     | 225    | 11    | 9     | 9     | 10    | 10    | 17    | 33    | 43    | 23    | 30   | 14    | 14    |
|                                                                                  | PrecMax     | 364    | 39    | 26    | 35    | 28    | 41    | 62    | 79    | 90    | 58    | 93   | 47    | 42    |
|                                                                                  | PrecMin     | 114    | 1     | 0     | 1     | 1     | 1     | 2     | 5     | 7     | 7     | 3    | 1     | 3     |
|                                                                                  | PrecStd     | 66     | 7     | 7     | 7     | 7     | 8     | 11    | 17    | 24    | 13    | 17   | 11    | 9     |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | -4.0  | -3.0  | -3.1  | 0.8   | 7.1   | 13.3  | 15.5  | 13.2  | 8.2   | 6.0  | 3.1   | 0.9   |
|                                                                                  |             | MCWB   | -6.4  | -5.5  | -5.3  | -1.9  | 3.0   | 6.8   | 7.7   | 6.8   | 4.0   | 2.5  | -0.1  | -1.3  |
|                                                                                  | 2%          | DB     | -8.5  | -8.7  | -8.2  | -2.2  | 4.4   | 10.6  | 13.3  | 11.7  | 6.1   | 1.6  | -1.5  | -3.5  |
|                                                                                  |             | MCWB   | -9.9  | -9.6  | -9.3  | -4.0  | 1.3   | 5.5   | 7.8   | 6.8   | 2.4   | -1.0 | -3.9  | -5.2  |
|                                                                                  | 5%          | DB     | -11.2 | -11.9 | -11.9 | -4.6  | 2.9   | 8.6   | 12.0  | 10.5  | 4.9   | -0.1 | -4.5  | -7.0  |
|                                                                                  |             | MCWB   | -12.2 | -12.7 | -12.9 | -6.2  | 0.4   | 4.7   | 7.6   | 6.5   | 1.7   | -2.3 | -6.4  | -8.2  |
|                                                                                  | 10%         | DB     | -14.0 | -14.8 | -14.4 | -6.5  | 1.7   | 6.9   | 10.9  | 9.2   | 3.9   | -1.6 | -6.7  | -9.6  |
|                                                                                  |             | MCWB   | -14.9 | -15.5 | -15.2 | -7.9  | -0.4  | 3.8   | 7.1   | 6.0   | 1.1   | -3.5 | -8.2  | -10.7 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | -6.4  | -5.7  | -5.3  | -1.7  | 3.5   | 7.6   | 9.7   | 8.5   | 4.3   | 2.5  | -0.3  | -1.4  |
|                                                                                  |             | MCDB   | -4.3  | -3.5  | -3.7  | 0.3   | 6.2   | 11.9  | 11.6  | 10.6  | 6.6   | 5.8  | 1.6   | 0.8   |
|                                                                                  | 2%          | WB     | -9.8  | -9.5  | -9.3  | -4.0  | 1.7   | 6.0   | 8.8   | 7.7   | 3.2   | -0.9 | -3.9  | -5.4  |
|                                                                                  |             | MCDB   | -8.6  | -8.6  | -8.3  | -2.5  | 3.6   | 9.5   | 11.6  | 9.7   | 5.2   | 0.9  | -1.6  | -3.5  |
|                                                                                  | 5%          | WB     | -12.3 | -12.7 | -12.9 | -6.1  | 0.7   | 4.9   | 8.1   | 7.2   | 2.2   | -2.1 | -6.3  | -8.2  |
|                                                                                  |             | MCDB   | -11.2 | -11.9 | -11.9 | -4.8  | 2.5   | 8.0   | 11.1  | 9.3   | 4.3   | -0.3 | -4.6  | -7.0  |
|                                                                                  | 10%         | WB     | -14.9 | -15.4 | -15.3 | -8.0  | -0.3  | 4.1   | 7.5   | 6.5   | 1.3   | -3.5 | -8.1  | -10.6 |
|                                                                                  |             | MCDB   | -14.0 | -14.8 | -14.5 | -6.6  | 1.4   | 6.5   | 10.3  | 8.7   | 3.4   | -1.8 | -6.8  | -9.5  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.5   | 5.9   | 7.0   | 8.3   | 6.7   | 5.1   | 5.1   | 4.2   | 3.9   | 3.7  | 4.3   | 5.0   |
|                                                                                  |             | MCDBR  | 8.5   | 9.8   | 9.4   | 9.9   | 8.0   | 7.8   | 7.2   | 6.1   | 5.1   | 4.7  | 6.9   | 7.4   |
|                                                                                  |             | MCWBR  | 7.4   | 8.8   | 8.4   | 8.1   | 5.7   | 4.4   | 3.6   | 3.3   | 3.4   | 3.6  | 5.9   | 6.4   |
|                                                                                  | 5% WB       | MCDBR  | 8.5   | 9.6   | 9.3   | 9.4   | 7.1   | 7.4   | 6.3   | 5.2   | 4.3   | 4.5  | 6.8   | 7.6   |
|                                                                                  |             | MCWBR  | 7.4   | 8.6   | 8.3   | 7.8   | 5.2   | 4.4   | 3.5   | 3.2   | 3.2   | 3.7  | 5.9   | 6.6   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | N/A    | 0.148 | 0.208 | 0.268 | 0.291 | 0.320 | 0.308 | 0.290 | 0.244 | 0.160 | N/A  | N/A   |       |
|                                                                                  | taud        | N/A    | 1.883 | 2.144 | 2.144 | 2.230 | 2.317 | 2.425 | 2.457 | 2.288 | 1.922 | N/A  | N/A   |       |
|                                                                                  | Ebn at Noon | 0      | 110   | 672   | 790   | 841   | 829   | 827   | 783   | 629   | 65    | 0    | 0     |       |
|                                                                                  | Edh at Noon | 0      | 6     | 47    | 86    | 97    | 94    | 81    | 65    | 43    | 4     | 0    | 0     |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.00   | 0.04  | 1.06  | 3.84  | 6.34  | 7.05  | 6.01  | 3.66  | 1.48  | 0.17  | 0.00 | 0.00  |       |
|                                                                                  | RadStd      | 0.00   | 0.00  | 0.05  | 0.16  | 0.28  | 0.45  | 0.34  | 0.32  | 0.11  | 0.01  | 0.00 | 0.00  |       |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (2 neighbors) | +0.98 | N/A     | N/A    | N/A     | N/A   | N/A   | -354        | -361    | N/A     | N/A     |

Nomenclature: See separate page